Practical 1 : RSA Encryption and Decryption Using Online Tool

Aim

To perform encryption and decryption using the RSA algorithm with the help of an online tool.

Theory

RSA algorithm is an asymmetric encryption technique that uses:

- **Public Key** → for encryption
- **Private Key** → for decryption

It ensures secure communication over networks.

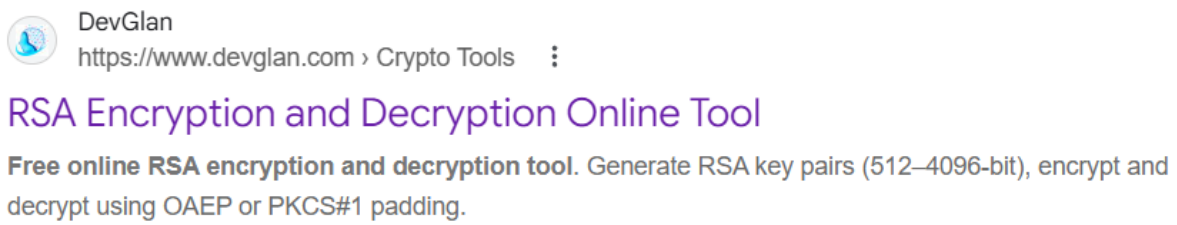
Tool Used

- RSA Encryption Tool (8gwifi.org)

Practical Steps

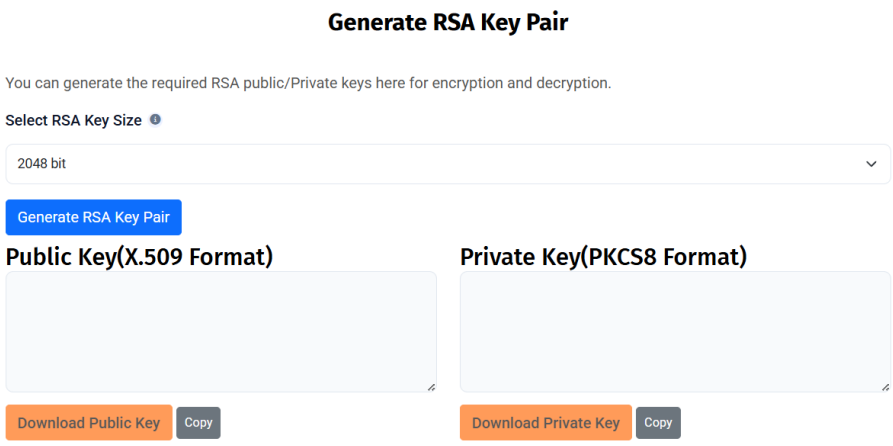
Step 1: Open the Tool

- Go to the RSA online tool website (8gwifi RSA tool)



Step 2: Generate Keys

- Click on **"Generate RSA Key Pair"**
- The tool will generate:
  - Public Key
  - Private Key



Step 3: Enter Plain Text

- In the input box, type your message
  Example: `Hello World`

Step 4: Encrypt the Message

- Paste the **Public Key**
- Click on **Encrypt**
- Copy the generated encrypted text (cipher text)

Step 5: Decrypt the Message

- Paste the encrypted text into the decryption box
- Paste the **Private Key**
- Click on **Decrypt**

Step 6: Verify Output

- The original message will be displayed after decryption

Output Example

- Plain Text: `Hello World`
- Encrypted Text: `a8F3kL9x...` (random encrypted data)
- Decrypted Text: `Hello World`

Result

The message was successfully encrypted using the public key and decrypted using the private key with RSA algorithm.

Conclusion

RSA provides high security by using two different keys. It is widely used in secure data transmission, digital signatures, and encryption systems.